

Impractical Turn-In System

AP Computer Science A — UML Object Class Relationship Diagram

CENTRAL APPLICATION DRIVER

App

ATTRIBUTES

- scanner : Scanner
- btnX : int
- btnY : int
- buttonCaught : boolean

METHODS

- + main(args : String[]) : void

ENCAPSULATED MODULE CLASSES (INTERACTING COMPONENTS)

Button (Sprint 1)

ATTRIBUTES

- x : int
- y : int
- LIMIT_X : int
- LIMIT_Y : int

METHODS

- + Button()
- + jump() : void
- + isClose(mX: int, mY: int) : boolean
- + getX() : int
- + getY() : int

Scrambler (Sprint 2)

ATTRIBUTES

- letters : char[]
- SIZE : int
- targetPhrase : String
- currentIndex : int

METHODS

- + Scrambler()
- randomize() : void
- + display() : void
- + checkInput(choice: int) : boolean
- + isComplete() : boolean

Obstacle (Sprint 3)

ATTRIBUTES

- errorMessage : String
- id : int

METHODS

- + Obstacle(msg: String, id: int)
- + getErrorMessage() : String

VerificationGrid (Sprint 4)

ATTRIBUTES

- grid : int[][]
- size : int
- isSorted : boolean

METHODS

- + VerificationGrid()

```
+ getId() : int
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+ displayGrid() : void  
+ selectionSortRow(row: int) : void  
+ checkPasscode() : boolean
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ARCHITECTURAL DEPENDENCY & INTERACTION MAPPING

App → Button: Instantiates and manages position boundary triggers (Sprint 1 dependency).

App → Scrambler: Triggers sequential letter validation loop via 1D Array indices (Sprint 2 dependency).

App → ArrayList<Obstacle>: Controls dynamic queue mutation, spawning and deleting objects (Sprint 3 dependency).

App → VerificationGrid: Resolves matrix organization state checks via nested loops (Sprint 4 dependency).